

Bedanta Bandyopadhyay

Aspiring Electronics Engineer | ML Enthusiast | CAD Modeler | Product Visualizer

Kolkata | vedantabandyopadhyay@gmail.com | +91 9038101994 | LinkedIn | GitHub | Portfolio Website | [Behance](#)

SUMMARY

Results-oriented engineer merging a robust background in automotive CGI with emerging expertise in Deep Learning and IoT. Experienced in managing end-to-end client projects and delivering high-impact visual solutions for diverse brands. Adept at rapid skill acquisition and passionate about building intelligent, connected systems

EDUCATION

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| 01. | Bachelor of Technology in Electronics and Communication Engineering
Institute of Engineering and Management, Newtown | <i>Expected Graduation date: 2027</i>
<i>CGPA: 8.86</i> |
| 02. | Higher Secondary (ISC)
Ram Mohan Mission High School | <i>Percentage: 73.2%</i> |
| 03. | Secondary (ICSE)
Ram Mohan Mission High School | <i>Percentage: 94.5%</i> |

TECHNICAL SKILLS

TOOLS AND SOFTWARES:

General: Microsoft Office | Git

Circuit and Embedded: Cadence Virtuoso | LT Spice | MATLAB | HFSS | Scaps1D | Arduino | Raspberry Pi | Jetson

Artificial Intelligence: Scikit-Learn | LangChain | RAG

Database Management: SQL | MongoDB

CAD Engineering: AutoCAD | SolidWorks | Fusion360

3D Modelling: Blender3D | Autodesk Maya | Autodesk 3DS Max

3D Rendering: Blender Cycles | Chaos Corona Renderer | Chaos V-Ray | Luxion Keyshot

Environment Design:

Post Production and Retouching: Adobe Photoshop | Blackmagic DaVinci Resolve | Blackmagic Fusion Design | Foundry Nuke

2D Illustrations: Adobe Illustrator

VFX: Adobe Premier Pro | Adobe After Effects

Texturing: Adobe Substance Painter | Adobe Substance Designer

Cloud and API Architecture: AWS | Flask

Web Development: HTML | CSS | ReactJS | ExpressJS | NodeJS

PROGRAMMING LANGUAGES: Python | C/C++ | Java | Rust | JavaScript

CORE COMPETENCIES: Circuit Design | Circuit Simulation | Fluid Simulation | UI-UX | CAD Modeling | Database Management Systems | Signal Analysis | Electronic Troubleshooting

ACADEMIC PROJECTS

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| 01. | Restro-Café (Full Stack WebDev): Designed a fully functional React Website for as a mockup for high profile restaurants, with a functional backend, Firebase and RazorPay integration for handling reservations, orders, deliveries and enquiries. | <i>May '26</i> |
| 02. | A.E.G.I.S. Core (Embedded Systems IoT Machine Learning): Developed a smart Women Safety device that works on Rpi 5 and Mediapipe to detect potential stalkers and disrupt any filming device(s) while transferring the location data of the victim to the concerned authorities and people. | <i>Feb '26</i> |
| 03. | Agro-Guard (IoT Embedded Systems): A smart agricultural monitoring system utilizing sensors and Arduino for data analysis for real-time crop health optimization. [PAPER] | <i>Jan '26</i> |
| 04. | Glaucoma Detection (Database Management Deep Learning): Developed a Deep Learning model that detects the Glaucoma Stage and the Risk Percentage using Fundus Scan Images. | <i>Dec '25</i> |
| 05. | 6T Static RAM Design (Cadence Virtuoso MATLAB): Design and simulation of a 6-Transistor Static Random-Access Memory cell, focusing on low-power consumption and noise margin stability. | <i>Nov '25</i> |
| 06. | 2 Switch Current Controlled Buck Boost Converter (Circuit Design Simulation): Engineered a high-efficiency power converter with current-control topology, designed for stable voltage regulation in variable load conditions. | <i>May '25</i> |

PREVIOUS EXPERIENCES

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| 01. | Automotive and Product Visualizer: (Freelance): Extensive 6-year tenure in remote freelance visualization. Specialized in hyper-accurate 3d design and lighting for automotive and product sectors. | <i>Apr '19 - Aug '25</i> |
| 02. | Co-Lead: (Ninja Fellowship): Led a team of 3d artists and worked alongside them to create striking visuals for various brands for over 2 years. | <i>Nov '22 - Mar '25</i> |

03.	3D Generalist and Environment Artist: (<i>The AutoFocus Studio</i>): Specializing in environment design and generalist work. Delivering high-fidelity visualizations and technical artwork for studio projects	<i>Aug '25 – Feb '26</i>
<u>INTERNSHIPS</u>		
01.	Internship on Advanced and Cleaner Materials in Solar Technology: Conducted performance optimization of Tri-Layer Lead-free Carbon-based Perovskite Solar Cells using Machine Learning and crafted a full detailed report on the entire process. [PAPER]	<i>Jun '25 - Jul '25</i>
<u>CERTIFICATIONS</u>		
01.	Business Intelligence and Analytics: (<i>NPTEL, IIT Madras</i>): Teaches about various machine learning algorithms that are used by companies for prediction of consumer response and sales of products. [VIEW]	<i>Jun '25</i>
02.	Joy of Computing with Python: (<i>NPTEL, IIT Ropar</i>): Teaches about the basics of python and goes up to the various advanced topics that form the foundation of data science and analysis. [VIEW]	<i>Nov '24</i>
03.	Cloud Computing: (<i>NPTEL, IIT Kharagpur</i>): Teaches about Cloud Computing and how various aspects and types of services are used and availed by the firms. [VIEW]	<i>Jun '25</i>
04.	Introduction to Internet of Things: (<i>Coursera, University of San Diego</i>): Teaches about the basics of Internet of Things and how we have arrived at the current technological standpoint in this field and demonstrates the use of Dragon board for various projects. [VIEW]	<i>Oct '24</i>
<u>EXTRA-CURRICULAR</u>		
01.	Graphics Lead: Iris The Photographic Society of UEMK	<i>Sep '24 - Dec '24</i>
02.	Media Lead: Iris The Photographic Society of UEMK	<i>Dec '24 - Aug '25</i>
03.	Tech Lead: Iris The Photographic Society of UEMK	<i>Dec '24 - Aug '25</i>
04.	Graphics Lead: Computer Society of India UEM Student Chapter	<i>Mar '25 - Jan '26</i>
05.	Manager: Computer Society of India UEM Student Chapter	<i>Aug '25 - Jan '26</i>
06.	Vice President: Computer Society of India UEM Student Chapter	<i>Jan '26 - Present</i>
07.	Class Representative	<i>Jan '26 - Present</i>